with this additional number zero became **Whole numbers.** The whole numbers can be visualized now as follows:



Fig 1.2

The system of whole numbers is denoted by $\mathbb{W}$.

Even zero was not sufficient to solve all problems. Think what happens when 4 is taken away from 6?

Draw a number line up to, say 9. Mark 6 by a dot on it.



Fig 1.3

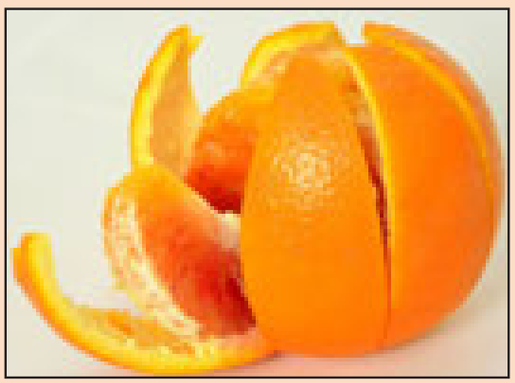
We know that, to subtract 4, we need to go 4 steps to the left side from 6. We will land on 2 and so the answer is 2. But what will happen if we want to subtract 6 from 4? This situation is where the humans needed (and created) negative numbers.



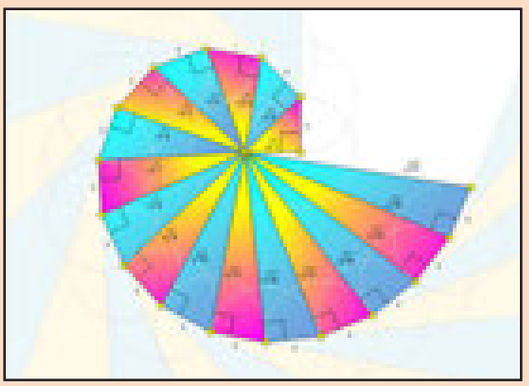
Fig 1.4

But, how can a number be negative? Simple! Just think of them as numbers less than zero. Including the negative integers with the whole numbers, we get the list of numbers called the **Integers.** The integers consist of zero, the natural numbers and the negatives of the natural numbers and it consists a list of numbers that stretch in either direction without end. The entire collection of integers is denoted by $\mathbb{Z}$.

MATHEMATICS ALIVE - NUMBERS IN REAL LIFE



If an orange is peeled off and 8 carpels are found, then one carpel represents the rational number $\frac{1}{8}$.



Finding the square root of numbers in the form spiral using the Pythagoras Theorem from Geometry

1.2 Rational Numbers

Even after coining integers, one could not relax! $10 \div 5$ is no doubt fine, giving the answer 2 but is $8 \div 5$ comfortable? Numbers between numbers are needed. $8 \div 5$ seen as 1.6, is a number between 1 and 2. But, where does $(-3) \div 4$ lie? Between 0 and –1. Similarly, where do you find

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